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UNIVERSITY
OF MIAMI



RELIABLE RENTALS

Conceptual Database Design | Part 1
DSC623 – Database Systems for Data Science | Spring 2026

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Case Study: Reliable Rentals

A relational database should be developed following the subsequent steps for the case study described below. All questions must be answered in the order described here; answers should be preceded by the question number. Answers must be **typed-in** and diagrams must be created using the **appropriate software** (e.g., Lucidchart), **no handwritten** parts will be accepted. You must **create a group** containing **3 people** and submit as a **group submission**.

Develop a conceptual data model reflecting the following requirements:

- Identify the main entities.
- Identify the main relationships between the entities identified in "a".
- Determine the multiplicity constraints for each relationship identified in "b".
- Identify attributes and associate them with entities or relationships.
- Determine candidate and primary key attributes for each (strong) entity.
- Generate the E-R diagram for the conceptual level (no FKs as attributes).

Notes: A report must be created including detailed documentation of each of the steps addressing all the required items. ER diagrams for the conceptual model must be included in the report. All assumptions made in the design must be clearly stated.

Case Study: Reliable Rentals

A company called Reliable Rentals rents out vehicles at different outlets (locations/sites). Each outlet has a number, address, phone number and a fax number. Each site is allocated a stock of vehicles for hire. The registration number uniquely identifies each vehicle for hire and is used when hiring a vehicle to a client.

Clients may hire vehicles for various periods of time. Each individual hire agreement between a client and the Company is uniquely identified using a hire number. Information stored on the vehicles for hire include: the vehicle registration number, model, make, engine size, capacity, current mileage, daily hire rate, and the current location (outlet) of each vehicle. The data stored on clients includes the client number (unique identifier), name (first and last name), home address, phone number, date of birth, and driving license number.

The data stored on a hire agreement includes the hire number, the client's number, name, address, and phone number, date the client started the hire period, date the client wishes to terminate the hire period, the vehicle registration number, model and make, the mileage before and after the hire period. Finally, information is stored on the staff based at various outlets including: staff number, name (first and last name), home address, home phone number, date of birth (DOB), sex, date joined the Company, job title, and salary. Each staff member is associated with a single outlet.

A. Main Entities

The following five entities were identified from the case study description:

- Outlet
- Vehicle

- Client
- Hire Agreement
- Staff

B. Main Relationships

The following relationships were identified between the entities:

Entity 1	Relationship	Entity 2
Outlet	stores	Vehicle
Vehicle	is rented through	Hire Agreement
Client	signs	Hire Agreement
Staff	works at	Outlet

C. Multiplicity Constraints

The following multiplicity constraints apply to each relationship:

Entity	Multiplicity	Relationship	Multiplicity	Entity
Outlet	1..1	stores	1..*	Vehicle
Vehicle	1..1	is rented through	0..*	Hire Agreement
Client	1..1	signs	0..*	Hire Agreement
Staff	1..*	works at	1..1	Outlet

Assumptions

- Each outlet stores one or more vehicles. Each vehicle is located at exactly one outlet at any given time.
- Each vehicle can appear in zero or more hire agreements over time. Each hire agreement involves exactly one vehicle.
- Each client may sign zero or more hire agreements. Each hire agreement is associated with exactly one client.
- Each staff member works at exactly one outlet. Each outlet employs one or more staff members

D. Attributes by Entity

Vehicle

- registration_number
- model
- make
- engine_size
- capacity
- current_mileage
- daily_hire_rate

Outlet

- outlet_number
- address
- phone_number
- fax_number

Client

- client_number
- name (composite)
 - first_name
 - last_name
- home_address
- phone_number
- date_of_birth
- driving_license_number

Hire Agreement

- hire_number
- start_date
- expected_end_date
- mileage_before
- mileage_after

Staff

- staff_number
- name (composite)
 - first_name
 - last_name
- home_address
- home_phone_number
- date_of_birth
- sex
- date_joined_company
- job_title
- salary

E. Candidate and Primary Keys

The table below identifies the candidate keys and selected primary key for each strong entity. Primary keys are underlined in the E-R diagram.

Entity	Candidate Key(s)	Primary Key
Vehicle	registration_number	registration_number
Outlet	outlet_number, address (assuming two outlets cannot exist in the same place/address)	outlet_number
Client	client_number, driving_license_number	client_number

Entity	Candidate Key(s)	Primary Key
Hire Agreement	hire_number	hire_number
Staff	staff_number	staff_number

F. Conceptual E-R Diagram

The E-R diagram below represents the conceptual data model for Reliable Rentals. Entities are shown with their attributes; primary keys are underlined. Composite attributes (name) are shown in italics with their sub-attributes indented. Foreign keys are intentionally omitted at the conceptual level.

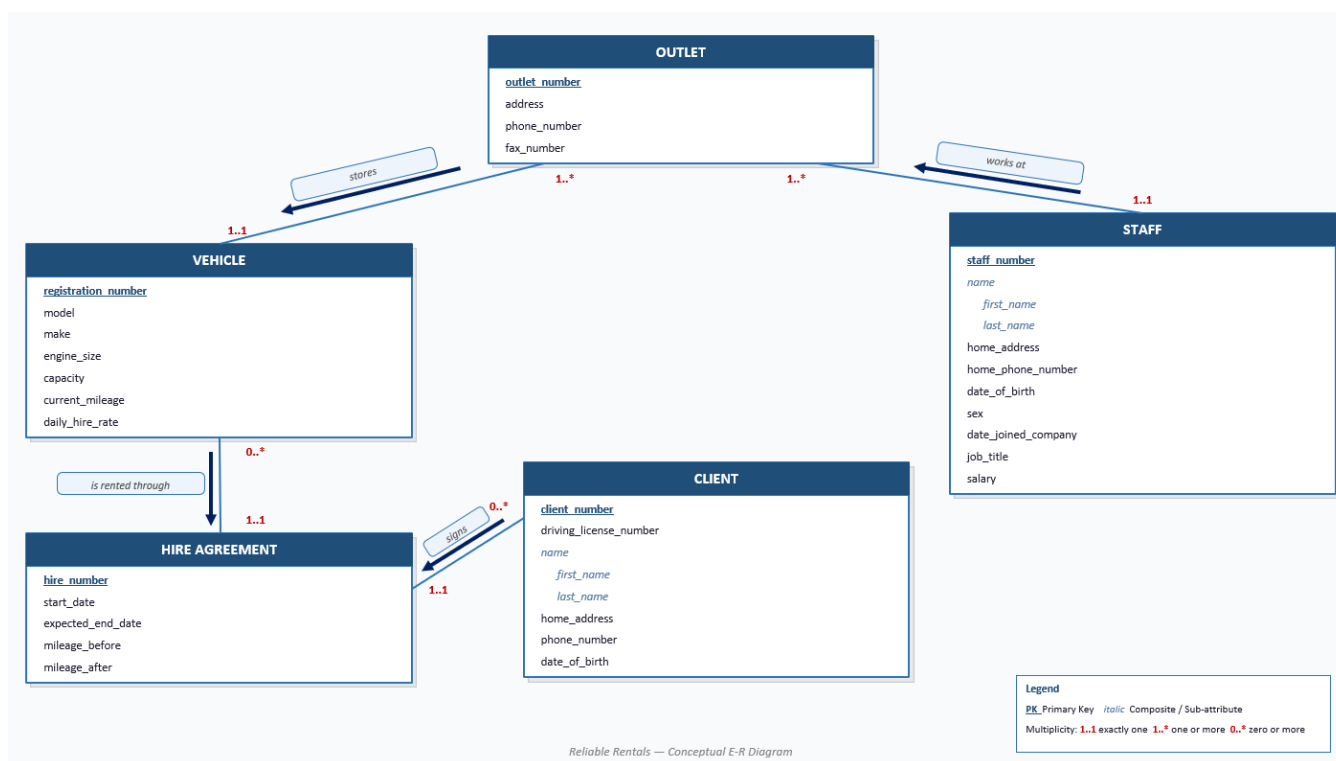


Figure 1. Conceptual E-R Diagram – Reliable Rentals